

- Q.20 Write an algorithm for insertion at star in doubly linked list. (CO2)
- Q.21 What is the limitation of a linear queue. How is it removed? (CO3)
- Q.22 What do you mean by traversal of a Tree? Explain in detail with example. (CO4)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)
- Q.23 Define data structure? Explain the various types of data structures in detail. (CO1)
- Q.24 Define an array? What are the types of array? Write an algorithm for addition of two matrices. (CO2)
- Q.25 Explain Quick sort technique in detail with algorithm and example. (CO6)

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2nd Sem / Artificial Intelligence & Machine Learning

Subject : Principles of Data Structure

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Quick sort is also known as _____ (CO5)
- a) Merge sort
 - b) Tree sort
 - c) Shell sort
 - d) Partition and exchange sort
- Q.2 Which data structure is used for implementing recursion? (CO3)
- a) Stack
 - b) Queue
 - c) List
 - d) Array
- Q.3 Which of the following is non-linear data structures? (CO1,4)
- a) Stacks
 - b) List
 - c) Strings
 - d) Trees

Q.4 Linked list is considered as an example of _____ type of memory allocation. (CO2)

- a) Dynamic b) Static
- c) Compile time d) Heap

Q.5 The operation of processing each element in the list is known as (CO1)

- a) Sorting b) Traversal
- c) Inserting d) Merging

Q.6 What are the disadvantages of arrays? (CO2)

- a) Index value of an array can be negative
- b) Elements are sequentially accessed
- c) There are chances of wastage of memory space if elements inserted in an array are lesser than the allocated size
- d) Data structure like queue or stack cannot be implemented

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define a pointer. (CO1)

Q.8 FIFO stands for _____. (CO3)

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Q.9 Linked list is a non-linear data structure.(True/False) (CO2)

Q.10 What do you mean by Siblings? (CO4)

Q.11 List two types of searching techniques. (CO5)

Q.12 What is two Dimensional Array? (CO2)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Differentiate between Top-Down Bottom-up design. (CO1)

Q.14 Write an algorithm for deletion from single dimensional array. (CO2)

Q.15 What are various applications of Linked list? (CO2)

Q.16 Explain Overflow and under flow condition in stack. (CO3)

Q.17 Sort the following list of elements using bubble sort. Show result after each step: (CO5)

16 15 12 19 18 50 17

Q.18 What is Binary Search Tree? Write down its properties. (CO4)

Q.19 Give difference between sequential search and binary search. (CO5)

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